Computer Science Research Document: Math, Algorithms, and Figures

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Abstract

This document demonstrates how to typeset mathematics, pseudocode, program source code, and figures in LaTeX.

1 Introduction and Math Notation

LaTeX provides robust tools for mathematical expressions. You can write inline equations like $f(n) = O(n \log n)$ or use display math:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n) \tag{1}$$

$$\int_0^\infty x \, \mathrm{d}x \tag{2}$$

By the Master Theorem, the solution to Equation (2) is $T(n) = \Theta(n \log n)$.

2 Algorithms

You can write structured pseudocode using the [algpseudocode](https://www.overleaf.com/learn/latex
package:

3 Computer Science Source Code

To include actual programming code, use the ‘listings‘ package:

Listing 1: Python implementation of Binary Search

```
def binary_search(A, t):
    low, high = 0, len(A) - 1
    while low <= high:
        mid = (low + high) // 2
        if A[mid] == t:
            return mid
```

Algorithm 1 Binary Search Algorithm

Require: Sorted array A of size n, target value t

Ensure: Index of t if found, else -1

$low \leftarrow 0$
$high \leftarrow n - 1$
while $low \leq high$ **do**
    $mid \leftarrow \lfloor (low + high)/2 \rfloor$
    if $A[mid] = t$ **then**
        return mid
    else if $A[mid] < t$ **then**
        $low \leftarrow mid + 1$
    else
        $high \leftarrow mid - 1$
    end if
end while
return -1

```
        elif A[mid] < t:
            low = mid + 1
        else:
            high = mid - 1
    return -1
```

4 Figures and Graphics

Figures are handled via the ‘graphicx‘ package. Place your image file (e.g., ‘diagram.png‘) in your project folder:

As shown in Figure 1, the data flows directly from the input layer to the processing node.

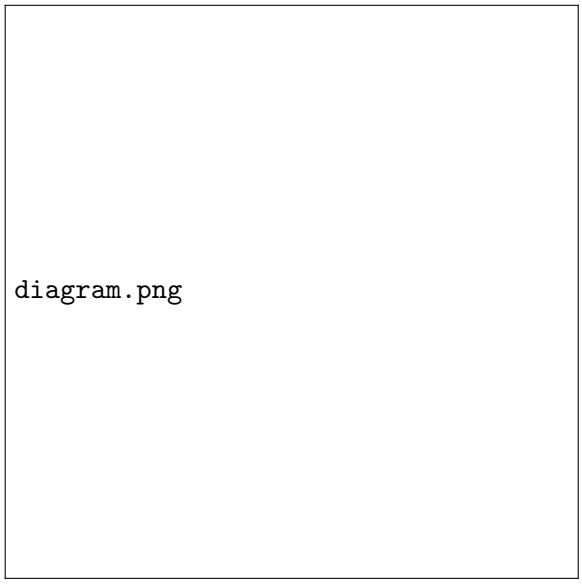


Figure 1: An architectural overview of the system flow.